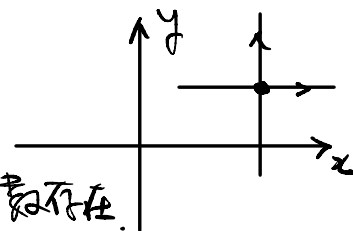


Recall: 偏导数;

可微 $\Rightarrow$ 偏导数存在;

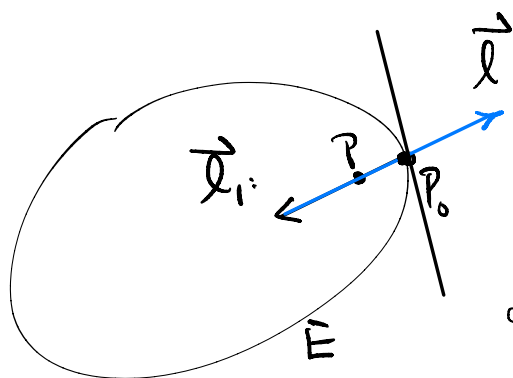
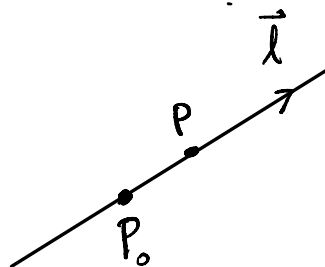
$\Rightarrow$ 对任意方向, 方向导数存在.



~~连续~~

$f(x, y, z)$: 在 $P_0(x_0, y_0, z_0)$ 处沿 $\vec{l}$ 的方向导数:

$$\lim_{P \rightarrow P_0} \frac{f(P) - f(P_0)}{|P - P_0|} = \frac{\partial f}{\partial \vec{l}} \Big|_{P_0}$$



$$\vec{l}_1 = -\vec{l}$$

$$\begin{aligned} \frac{\partial f}{\partial \vec{l}} \Big|_{P_0} &= \lim_{P \rightarrow P_0} \frac{f(P) - f(P_0)}{|P - P_0|} \\ &\stackrel{\text{定义}}{=} - \frac{\partial f}{\partial \vec{l}_1} \Big|_{P_0} \end{aligned} \quad \underbrace{\vec{P_0P} \text{ 与 } \vec{l} \text{ 一致}}$$

如果 $\vec{l}$ 是 ∂E 的外法线方向:

$P_0 \in \partial E$:

$\frac{\partial f}{\partial \vec{l}} \Big|_{P_0}$ 称为边界上外法线方向的方向导数.

方向导数的计算:

梯度: $\nabla f(P_0) = \{f_x(P_0), f_y(P_0), f_z(P_0)\}$.

$$\begin{aligned} \frac{\partial f}{\partial \vec{l}} \Big|_{P_0} &= \nabla f(P_0) \cdot \vec{l}_0 \\ &= \|\nabla f(P_0)\| \cdot \underbrace{\|\vec{l}_0\|}_{=1} \cos \theta \end{aligned}$$

$\vec{l}_0 = \frac{\vec{l}}{|\vec{l}|}$, 单位向量.

θ : $\vec{l}$ 与 $\nabla f(P_0)$ 之夹角.

$$= \|\nabla f(P_0)\| \cdot \cos \theta = \sqrt{(f_x(P_0))^2 + (f_y(P_0))^2 + (f_z(P_0))^2} \cos \theta$$

$\cos\theta = 1$ i.e. $\theta = 0$. $\vec{l}$ 与 $\nabla f(p_0)$ 相同. $\cos\theta$ 最大.
 此时 $\frac{\partial f}{\partial l}|_{p_0}$ 最大.

$$\nabla f(p_0) = \{f_x(p_0), f_y(p_0), f_z(p_0)\}$$

二元函数 $z = f(x, y)$: 在 $P_0(x_0, y_0)$ 处梯度

$$\nabla f(x_0, y_0) = \{f_x(x_0, y_0), f_y(x_0, y_0)\}$$

$z = f(x, y)$: 曲面; 若 $f(x, y)$ 可微:

$$dz = f_x dx + f_y dy \quad \text{—— 微分;}$$

取: $x_0 \rightarrow x = x_0 + \Delta x$, $y_0 \rightarrow y = y_0 + \Delta y$:

$$\Delta z = f(x, y) - f(x_0, y_0) = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

$$= f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + o(\rho), \quad (\rho = \sqrt{(\Delta x)^2 + (\Delta y)^2})$$

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \quad \leftarrow \text{切平面;}$$

$$\text{点 } (x_0, y_0, z_0) \text{ (在曲面上); } \quad z_0 = f(x_0, y_0).$$

切平面;

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0.$$

法方向: $\vec{n} \triangleq (f_x(x_0, y_0), f_y(x_0, y_0), -1)$.

(normal vector: 法向量).

曲面 $z = f(x, y)$: 由隐函数决定: $F(x, y, z) = 0$.
 ($F_z \neq 0$)

$$F(x, y, f(x, y)) \equiv 0;$$

$$F_x + F_z \cdot f_x = 0 \Rightarrow f_x = -\frac{F_x}{F_z}; \quad \checkmark$$

$$F_y + F_z \cdot f_y = 0 \Rightarrow f_y = -\frac{F_y}{F_z}; \quad \checkmark$$

$$\Rightarrow \text{法方向 } \vec{n} = \{f_x, f_y, -1\}$$

$$= \left\{ -\frac{F_x}{F_z}, -\frac{F_y}{F_z}, -1 \right\}$$

$$= -\frac{1}{F_z} \{F_x, F_y, F_z\}$$

$$\vec{n} \parallel \{F_x, F_y, F_z\}.$$

证: $F(x, y, z)$ 可微: $\Rightarrow$ 曲面 $F(x, y, z) = 0$: 法向量为

$$\text{就是 } \nabla F = \{F_x, F_y, F_z\}.$$

曲面光滑:

$$\textcircled{1} \quad \nabla F(x, y, z) \neq \vec{0};$$

$$\textcircled{2} \quad F_x, F_y, F_z \text{ 连续}.$$

$$\Leftrightarrow F(x, y, z) = 0: \quad F(x, y, z) \text{ 连续可微且 } \nabla F \neq \vec{0}.$$

§15.4. Taylor 公式.

Recall 一元函数 Taylor 公式.

$$f(x) = P_n(x) + R_n(x), \quad x_0:$$

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

—— Taylor 公式

$R_n(x)$. 余项.

$$x-x_0 \rightarrow 0.$$

① $R_n(x) = o((x-x_0)^n)$ ———— Peano型余项.

要求 $f(x)$ 有 $n-1$ 阶导数; $f(x)$ 在 x_0 有 n 阶导数.

$$(n=1: f(x) \text{ 在 } x_0 \text{ 可导} \Rightarrow f(x) = f(x_0) + f'(x_0)(x-x_0) + o(x-x_0))$$

② $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1}$ ———— Lagrange型余项.

ξ 介于 x_0 与 x 之间;

要求 $f(x)$ 有 $n+1$ 阶导数

③ 积分型余项: ———— Cauchy型余项.

?

($n=1$: $f(x)$ 有二阶导数: $\Rightarrow$

$$f(x) - f(x_0) = \int_{x_0}^x f'(s) ds$$

$$= \left(\int_0^1 f'(x_0 + t(x-x_0)) dt \right) (x-x_0)$$

$$\triangle S = x_0 + t(x-x_0)$$

$$ds = (x-x_0) dt,$$

$$s: x_0 \rightarrow x$$

$$t: 0 \rightarrow 1.$$

$f(x)$ 有二阶导数:

$$= \int_0^1 f'(x_0 + t(x-x_0)) d(1-t) \cdot (x-x_0)$$

$$= -(1-t)f'(x_0 + t(x-x_0)) \Big|_0^1 \cdot (x-x_0)$$

$$+ \int_0^1 (1-t)f''(x_0 + t(x-x_0)) dt \cdot (x-x_0)^2$$

$$f(x) - f(x_0) = f'(x_0)(x-x_0) + \int_0^1 (1-t)f''(x_0 + t(x-x_0)) dt \cdot (x-x_0)^2$$

对于多元函数,

设 $f(x, y)$ 在 (a, b) 邻域中有 $n+1$ 阶连续偏导数.

$$(x, y) = (a+h, b+k), \quad \Delta x = h, \quad \Delta y = k.$$

$$f(x, y) - f(a, b) = f(a+th, b+tk) \Big|_{t=0}^{t=1} = \varphi(1) - \varphi(0)$$

$\varphi(t) \triangleq f(a+th, b+tk)$: 是 t 的 $n+1$ 阶连续可导函数.

$$\varphi(t) = \varphi(0) + \varphi'(0)t + \frac{\varphi''(0)}{2!}t^2 + \dots + \frac{\varphi^{(n)}(0)}{n!}t^n + \frac{\varphi^{(n+1)}(\xi)}{(n+1)!}t^{n+1}$$

$$t=1.$$

ξ 介于 0, 1 之间

$$\varphi(1) - \varphi(0) = \varphi'(0) + \frac{\varphi''(0)}{2!} + \dots + \frac{\varphi^{(n)}(0)}{n!} + \frac{\varphi^{(n+1)}(\xi)}{(n+1)!}$$

$$\varphi'(t) = f_x(a+th, b+tk) \cdot h + f_y(a+th, b+tk) \cdot k$$

$$t=0 \Rightarrow$$

$$\begin{aligned} \varphi'(0) &= f_x(a, b) \cdot h + f_y(a, b) \cdot k = \{f_x(a, b), f_y(a, b)\} \cdot \{h, k\} \\ &= \nabla f(a, b) \cdot (h, k) \end{aligned}$$

$$\begin{aligned} \varphi''(t) &= f_{xx}(a+th, b+tk) \cdot h^2 + f_{xy}(a+th, b+tk) \cdot kh \\ &\quad + f_{yx}(a+th, b+tk) \cdot hk + f_{yy}(a+th, b+tk) \cdot k^2 \end{aligned}$$

$$\begin{aligned} &= f_{xx}(a+th, b+tk) h^2 + 2f_{xy}(a+th, b+tk) \cdot hk \\ &\quad + f_{yy}(a+th, b+tk) k^2 \end{aligned}$$

$$\varphi''(0) = f_{xx}(a, b) h^2 + 2f_{xy}(a, b) hk + f_{yy}(a, b) k^2$$

$$= (h, k) \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} = (h, k) \cdot H \cdot \begin{pmatrix} h \\ k \end{pmatrix}.$$

$$H \triangleq \begin{pmatrix} f_{xx}(a,b) & f_{xy}(a,b) \\ f_{xy}(a,b) & f_{yy}(a,b) \end{pmatrix} \text{ ————— 称为 Hesse 矩阵}.$$

$$\dots \dots \varphi'''(t), \varphi'''(0), \dots$$

$$\begin{aligned} f(x,y) - f(a,b) &= \varphi(1) - \varphi(0) \\ &= \varphi'(0) + \frac{\varphi''(0)}{2!} + \dots + \frac{\varphi^{(n)}(0)}{n!} + \frac{\varphi^{(n+1)}(\theta)}{(n+1)!} \\ &= \nabla f(a,b) \cdot (h,k) + \frac{1}{2!} (h,k) \cdot H \cdot \begin{pmatrix} h \\ k \end{pmatrix} \\ &\quad + \dots \quad 0 < \theta < 1. \end{aligned}$$

用归纳法:

$$\begin{aligned} \nabla f(a,b) \cdot (h,k) &= \{f_x(a,b), f_y(a,b)\} \cdot (h,k) \\ &= h \partial_x f(a,b) + k \partial_y f(a,b) \\ &\triangleq (h \partial_x + k \partial_y) f(a,b). \end{aligned}$$

$$L_1 = \underline{(h \partial_x + k \partial_y)}$$

$$L_1 f(x,y) = h \partial_x f(x,y) + k \partial_y f(x,y)$$

$$\begin{aligned} (h,k) \cdot H \cdot \begin{pmatrix} h \\ k \end{pmatrix} &= f_{xx}(a,b) h^2 + 2 f_{xy}(a,b) h k + f_{yy}(a,b) k^2 \\ &= h^2 \partial_x^2 f(a,b) + 2 h k \partial_{xy}^2 f(a,b) + k^2 \partial_y^2 f(a,b). \end{aligned}$$

$$L_2 = (h \partial_x + k \partial_y)^2 = h^2 \partial_x^2 + 2 h k \partial_{xy} + k^2 \partial_y^2:$$

$$L_2 f(x,y) = h^2 \partial_x^2 f(x,y) + 2 h k \partial_{xy}^2 f(x,y) + k^2 \partial_y^2 f(x,y)$$

$$L_2 f(a,b) = (h,k) H \begin{pmatrix} h \\ k \end{pmatrix}.$$

$$\varphi(t) = f(a+th, b+tk): \quad \text{用归纳法:}$$

$$\varphi'(t) = f_x \cdot h + f_y \cdot k = (h\partial_x + k\partial_y) f(a+th, b+tk).$$

$$\varphi''(t) = (h\partial_x + k\partial_y)^2 f(a+th, b+tk)$$

.....

$$\varphi^{(n)}(t) = (h\partial_x + k\partial_y)^n f(a+th, b+tk).$$

$$\varphi^{(n)}(0) = (h\partial_x + k\partial_y)^n f(a, b).$$

$$\varphi^{(n+1)}(\xi) = (h\partial_x + k\partial_y)^{n+1} f(a+\xi h, b+\xi k).$$

$$\begin{aligned} f(x, y) - f(a, b) &= \varphi(1) - \varphi(0) \\ &= \varphi'(0) + \frac{\varphi''(0)}{2!} + \dots + \frac{\varphi^{(n)}(0)}{n!} + \frac{\varphi^{(n+1)}(\xi)}{(n+1)!} \\ &= (h\partial_x + k\partial_y) f(a, b) + \frac{1}{2!} (h\partial_x + k\partial_y)^2 f(a, b) \\ &\quad + \dots + \frac{1}{n!} (h\partial_x + k\partial_y)^n f(a, b) \\ &\quad + \frac{1}{(n+1)!} (h\partial_x + k\partial_y)^{n+1} f(a+\theta h, b+\theta k). \end{aligned}$$

$$\text{—————} = \text{Taylor 公式} \quad 0 < \theta < 1.$$

$$f(x, y) = P_n(h, k) + R_n(h, k).$$

$$\begin{aligned} P_n(h, k) &= f(a, b) + (h\partial_x + k\partial_y) f(a, b) + \frac{1}{2!} (h\partial_x + k\partial_y)^2 f(a, b) + \dots \\ &\quad + \frac{1}{n!} (h\partial_x + k\partial_y)^n f(a, b). \end{aligned}$$

$$\text{—————} f(x, y) \text{ 在 } (a, b) \text{ 处 } n \text{ 阶 Taylor 公式}$$

$$R_n(h, k) = \frac{1}{(n+1)!} (h\partial_x + k\partial_y)^{n+1} f(a+\theta h, b+\theta k). \quad \text{—— 余项.}$$

一阶展开式: $L_1 = h\partial_x + k\partial_y$: —— 微分算子:

$$f(x, y) = f(a, b) + \underbrace{(h\partial_x + k\partial_y)}_{\text{一阶展开}} f(a, b) + o(\sqrt{h^2 + k^2}).$$

$$L_1 f(a, b) = h\partial_x f(a, b) + k\partial_y f(a, b) = \nabla f(a, b) \cdot (h, k)$$

$$f(x, y) = f(a, b) + f_x(a, b)h + f_y(a, b)k + o(\rho) \quad \begin{array}{l} h = x - a \\ k = y - b \\ \rho = \sqrt{h^2 + k^2} \end{array}$$

二阶展开式:

$$L_2 = (h\partial_x + k\partial_y)^2 = h^2\partial_x^2 + 2hk\partial_x\partial_y + k^2\partial_y^2. \quad \text{二阶微分算子}$$

$$\begin{aligned} L_2 f(a, b) &= h^2\partial_x^2 f(a, b) + 2hk\partial_x\partial_y f(a, b) + k^2\partial_y^2 f(a, b) \\ &= f_{xx}(a, b)h^2 + 2f_{xy}(a, b)hk + f_{yy}(a, b)k^2. \end{aligned}$$

$$\begin{aligned} f(x, y) &= f(a, b) + f_x(a, b)h + f_y(a, b)k \\ &\quad + \frac{1}{2!} \left(f_{xx}(a, b)h^2 + 2f_{xy}(a, b)hk + f_{yy}(a, b)k^2 \right) \\ &\quad + o(\rho^2). \end{aligned}$$

$\uparrow$
 $h, k \sim \text{二次型}$

$$\begin{aligned} &= f(a, b) + \nabla f(a, b) \cdot (h, k) \\ &\quad + \frac{1}{2!} (h, k) \cdot H \cdot \begin{pmatrix} h \\ k \end{pmatrix} + o(\rho^2), \end{aligned}$$

$$\text{Hesse 矩阵: } H = \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{xy}(a, b) & f_{yy}(a, b) \end{pmatrix}.$$

作业. P. 139, 1, 2.